

Course Code	EC24P
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Class	T.E – EXTC. SEM-V
Division	B
Date of Performance	18-7-25
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EXPERIMENT NO.1

Estimated time to complete this experiment: 02 hours.

Objective: To simulate and Plot the Transfer and output characteristics of 180nm N channel MOSFET using CADENCE VIRTUOSO

CO to be achieved: CO1.

Expected Outcome of Experiment:

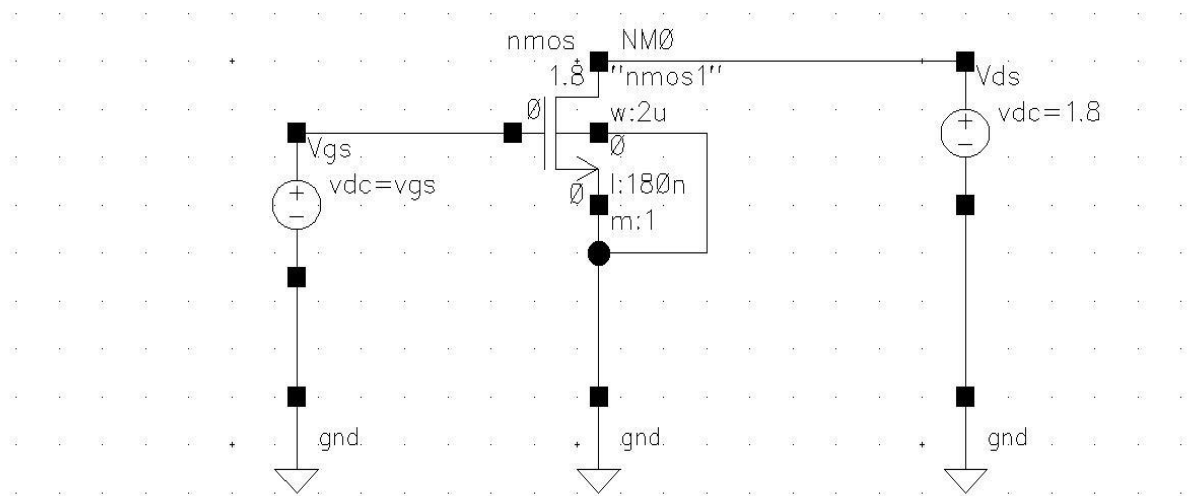
- Students will learn cadence virtuoso.
- To plot output and transfer characteristics of MOSFET.
- To calculate small signal parameters of MOSFET.

Pre Lab/ Prior Concept: Knowledge of working of MOSFET.

Apparatus/TOOL:

- PC
- CADENCE VIRTUOSO.

Circuit Diagram:



Procedure.

- Open the cadence virtuoso tool in Red-Hat Linux. The commands are
 - `ssh`
 - `source /home/install/cshrc`
 - `virtuoso.`
- Create the Library and cell view

- Open log window using terminal.
- Click on the new option, now create library with gpdk180 from technology library.
- Again click the on new option, and set the cell view.
- Cadence Virtuoso dashboard will be opened.
- For characterization of 180nm NMOS Transistor
- Select NMOS transistor from add instance and select its channel length as 180nm (Nano meter) and channel width as 2u(micro-meter).
- Construct the circuit shown in circuit diagram in Cadence.
- To plot Transfer Characteristics, keep VDD=1.8V and vary the VGG from 0 to 1.8V
- From the transfer characteristics calculate gm of given NMOS Transistor.

$$gm = \frac{ID2-ID1}{VGS2-VGS1} \text{ } \cup$$

- To plot output or drain to source characteristics keep VGG=0.8V and vary the VDD from 0 to 1.8V.
- From the output characteristics calculate ro for the given transistor.

$$ro = \frac{VDS2 - VDS1}{ID2 - ID1} \Omega$$

- Note down the value of drain current in saturation region and using that calculate the value of process Trans conductance $\mu nCOX$, using the drain current equation.

$$ID = \frac{\mu nCOX}{2} * \left(\frac{W}{L}\right) * (VGS - VTN)^2$$

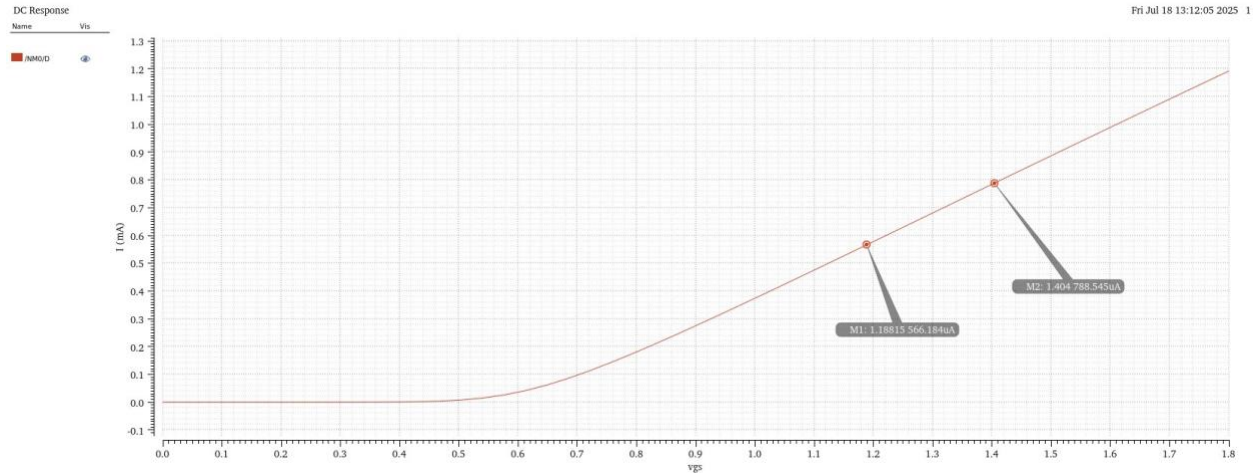
- Using the value of ro Calculate the value of λ (Channel length modulation coefficient) by using equation.

$$ro = \frac{1}{\lambda * ID} \Omega$$

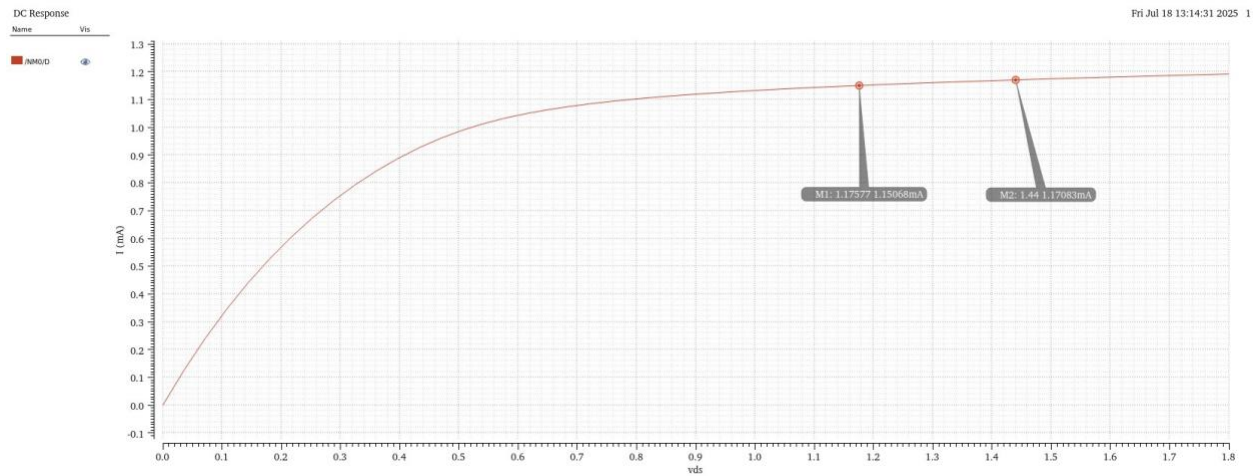
- Finally calculate the Intrinsic gain of MOSFET= gm * Ro
- Then Plot the Transfer characteristics for following values of VDS
VDS=1.8V, VDS=1.5V, VDS=1V
- Output characteristics for following values of VGS.
VGS=1.8V, VGS=1.5V, VGS=1V.

Results:

Transfer Characteristics:



Output Characteristics:



Observation Table.

VDS=1.8V	
VGS VARIYING	
gm	1.00481m
$\mu_n C_{ox}$	0.00025779
VGS=1V	
VDS VARIYING	
ro	27729.38443
λ	0.100720

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signal OP("/NM0" "??")

Iavl NaN
beff 3.57945m
betaeff 5.59158m
cbb 2.62628f
cbd -1.62798f
cbdbi -1.50322a
cbg -499.874a
cbs -498.426a
cbsbi -498.426a
cdb -1.62663f
cdd 2.35931f
cddb1 225.668z
cdg -733.108a
cbs 428.67z
cgb -474.283a
cgbovl 26.5891a
cgd -725.772a
cgdbi 6.82765a
cgdovl 732.6a
cgg 3.72029f
cggbi 2.2285f
cgs -2.52024f
cgsbi -1.78764f
cgsovl 732.6a
cjd 1.62648f
cjs 0
cm NaN
cmb NaN
cmx NaN
co NaN
csb -525.374a
csd -5.5501a
csg -2.48731f
css 3.01823f
cssbi 2.28563f
ctype 2.14748G
dtsh NaN
fqslim NaN
ft NaN
fug 42.9861G
gbd 813.616n
gbs 0
gds 32.1041u
gm 1.00481m
gmb 292.149u
gmbs 292.149u
gmoveri NaN
gmoverid 2.68485
go NaN
i1 374.399u
i3 -374.253u
i4 -146.812n
ib NaN
ibd -146.812n
ibe -146.814n
ibs 0
ibulk -146.812n
id 374.399u
idb 129.909a
ide 374.399u
ids 374.253u
ig NaN
igb 0

igcd 0
igcs 0
igd 0
ige 0
igidl 0
igisl 0
igs 0
is -374.253u
isb -0
ise 374.399u
isub 146.812n
opdef 2.14748G
pwr 673.919u
qb -2.22695f
qbd -3.49249f
qbi -2.20037f
qbs 0
qd 585.896a
qdi -184.086z
qg 3.17624f
qgi 3.00313f
qinv 2.73758m
qsi -802.582a
qsrco -1.53518f
region 2
reversed 0
ron 4.80958K
rout 31.1487K
sdint 2.14748G
sdop -1
self_gain 31.2985
taul NaN
tk NaN
trise NaN
ueff 45.7978m
vbs 0
vdb 1.8
vds 1.8
vdsat 250.77m
vdsat_marg NaN
vdss 250.77m
vearly 11.6621
vfbeff -1.11772
vgb 1
vgd -800m
vgs 1
vgsteff 457.32m
vgt 457.377m
vsat_marg 1.54923
vsb -0
vth 542.623m
vth0 NaN
vth_drive NaN

Calculation:

Calculations:

1) $V_{TH} = 0.5 \text{ V}$

$$2) g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{I_{D2} - I_{D1}}{V_{GS2} - V_{GS1}}$$

$$= \frac{867.191 \times 10^{-6} - 687.191 \times 10^{-6}}{1.4805 - 1.3063}$$

$$= 0.001033794$$

$$3) r_o = \frac{\Delta V_{DS}}{\Delta I_D} = \frac{1.4150 - 1.224}{361.499 \times 10^{-6} - 354.611 \times 10^{-6}}$$

$$= 27729.38443$$

$$4) I_{D(sat)} = \frac{I_{D1} + I_{D2}}{2} = \frac{361.499 \times 10^{-6} + 354.611 \times 10^{-6}}{2}$$

$$= 0.00035805$$

$$I_{D(sat)} = \frac{U_n C_{ox}}{2} \left(\frac{W}{L} \right)^2 [V_{GS} - V_{th}]^2$$

$$0.00035805 = \frac{U_n C_{ox}}{2} \left(\frac{2 \mu m}{180 nm} \right)^2 [1 - 0.5]^2$$

$$U_n C_{ox} = 0.000257796$$

$$5) r_o = \frac{1}{\lambda I_{D(sat)}}$$

$$\lambda = \frac{1}{r_o \times I_{D(sat)}} = 0.100720$$

(channel length modulation coefficient)

$$6) g_{mTo} \text{ (Intrinsic gain)}$$

$$0.001033794 \times 27729.38443$$

$$= 0.000286443$$

$$= 0.000286443$$

$$0.001033794 \times 27729.38443$$

$$= 28.6664$$

Conclusion:

In this experiment, you successfully simulated and analyzed the transfer and output characteristics of a 180nm N-channel MOSFET using Cadence Virtuoso. By observing the variation of gate and drain voltages, you calculated important parameters like transconductance, channel length modulation coefficient, and intrinsic gain. You also understood how the MOSFET operates as a voltage-controlled device, where the gate voltage controls the current flow between source and drain through the electric field effect.

Post lab Questions.

- **Why MOSFET is known as Field effect transistor?**

MOSFET stands for Metal-Oxide-Semiconductor Field Effect Transistor. It is called a "Field Effect Transistor" because the current flowing between two terminals (source and drain) is controlled by an electric field applied to the gate terminal. This field changes how easily current flows, letting the device switch or amplify signals. Instead of using current to control current (like in Bipolar Junction Transistors), MOSFETs use voltage at the gate to control the flow of current, making it a voltage-controlled device

- **Define threshold voltage of MOSFET.**

The threshold voltage of a MOSFET (often written as V_{th} or $V_{GS(th)}$) is the minimum voltage that must be applied to the gate terminal for a channel to form between source and drain, allowing current to start flowing. If the voltage at the gate is less than the threshold voltage, the MOSFET remains OFF (no current flows); if it's more than the threshold voltage, the MOSFET turns ON (current flows)

- **What is the maximum supply voltage for 180nm MOSFET?**

For a 180nm MOSFET process, the typical maximum supply voltage is around 1.8V. This means you should not exceed 1.8V when using circuits designed on the 180nm technology node. Using a higher voltage can damage the device or reduce its life

- **Give the difference between NMOS & PMOS in tabular form**

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Parameter	NMOS	PMOS
Carrier type	Electrons (negative)	Holes (positive)
Operation	ON when Gate > Source (positive voltage)	ON when Gate < Source (negative voltage)
Symbol	Arrow towards gate	Arrow away from gate
Gate voltage (for ON)	Positive	Negative
Switching speed	Faster	Slower
Power consumption (ON)	Lower	Higher
Noise immunity	Less	More
Common use cases	Digital logic, amplifiers	Level translators, analog circuits

Practical Performance (5 Marks)	Presentation and Conclusion (5 Marks)	Total (10 Marks)	Sign	Date of Submission.